

# Aime Cesaire

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## SUMMARY

IT Systems Engineer with a robust foundation in **AI implementation, systems, data, and full-stack development**. Proven experience in bridging cloud, data, and generative AI engineering to drive revenue and efficiency. A technically rigorous problem-solver who has built systems and automations **reclaiming \$120K+/yr in losses and unlocking \$100K+/yr in revenue**. Dedicated to building scalable, secure, and data-driven infrastructure in fast-paced, innovation-first environments.

## EDUCATION

**B.A. with Distinction in Computer Science & Math, GPA: 3.95** | *Amherst College* | Amherst, MA

*Aug 2021 - May 2025*

- Relevant Coursework: Natural Language Processing, Artificial Intelligence, Machine Learning, Nonlinear Optimization, Probability and Computing, Computer Security, Computer Systems, Computer Networks, Data Structures, Algorithms
- Activities: Association for Computing Machinery, Artificial Intelligence in the Liberal Arts

**Study Abroad Semester, GPA: 3.67** | *Sorbonne Sciences & Nanterre Université* | Paris, France

*Jan 2024 - May 2024*

Relevant Coursework: Cryptography

## TECHNICAL SKILLS

- **AI & Data:** LangChain, PyTorch, Hugging Face Transformers, RAG, Pandas, SQL, Postgres, MongoDB
- **Full stack:** Python, TypeScript, Rust, C++, Node.js, React, React Native, Flask, REST, GraphQL, Git, CI/CD
- **Cloud & Systems:** AWS, Azure, Google Cloud, PowerShell, Power Automate, Docker, Linux, IAM & RBAC, DHCP & DNS, Postman

## PROFESSIONAL EXPERIENCE

**Systems Engineer** | *Gratz Pilates* | Philadelphia, PA

*Jun 2025 - Present*

- **Technical Lead** for a full-stack Shopify-HubSpot integration, coordinating with the external development team and defining **system architecture** via UML modeling. Managed the full **SDLC**, from requirements gathering to code reviews, ensuring seamless data synchronization across the enterprise.
- Developed and maintained a suite of specialized **Python and PowerShell** scripts to automate backend service monitoring, **API testing**, and critical data backups. Executed **complex SQL** queries across sales and manufacturing systems to ensure high-fidelity reporting and data integrity.
- Manage and optimize a **hybrid cloud** environment spanning **AWS, Azure**, Digital Ocean, on-premises Linux (Apache), and Windows servers. Orchestrate the deployment of **scalable infrastructure**, including **containerized environments**, network resources, and high-availability databases.
- Architected a comprehensive security framework across on-premises, PaaS, and SaaS environments. Hardened system integrity by implementing enterprise-grade **IAM and RBAC** protocols, while deploying integrated solutions to secure network endpoints and physical infrastructure.
- Resolved a critical HubSpot integration bug in Fishbowl Inventory by **reverse-engineering JAR files** and performing root-cause analysis; authored a **technical proposal and code fix** that was successfully adopted by the Fishbowl development team.
- Optimized **AI agents** within Microsoft 365, Shopify, and HubSpot ecosystems to deliver high-fidelity, data-driven customer support. Leveraged **context engineering** and **RAG** to reduce hallucinations and ensure AI-generated responses aligned with internal workflows and product specifications.

**IT Support Engineer** | *Amherst College* | Amherst, MA

*Sep 2022 - May 2025*

- Performed **20+ root-cause investigations into major software outages**, utilizing **Postman, Chrome dev tools**, and other diagnostic tools to debug SaaS APIs and minimize service disruptions for critical enterprise systems.
- Managed the end-to-end lifecycle and configuration of hardware and software across distributed networks, deploying **JAMF Pro** and **Microsoft Intune** to automate device imaging, security enrollment, and standardized configuration management across multiple physical locations.
- Automated ticket triage and resolution workflows for **3K+ technical escalations**, **reducing average response time by 50%** while collaborating with senior engineering teams to debug and resolve complex system-level issues.
- Facilitated the enterprise-wide rollout of MFA, leading to an **80% reduction in security incidents** and spearheading independent network surveys to optimize on-site connectivity and hardware reliability.

**Program Assistant for AI in the Liberal Arts** | *Amherst College* | Amherst, MA

*Sep 2022 - May 2023*

- Researched and prototyped emerging AI architectures, including **LLMs, Diffusion Models, GANs**, and **Multi-modal Neural Networks**, to evaluate technical feasibility and inform the strategic evolution of organizational programs.
- Co-authored two professional websites and authored **50+ article summaries on AI architectures**, while coordinating six expert-led panels with **5K+ attendees** to drive a **3x increase in membership**.

## TECHNICAL PROJECTS

**RAG-based Agentic Workflows** | *LLMs, RAG, Python, AI Agent Orchestration*

- Architected **autonomous AI agents** within Microsoft 365, Shopify, and HubSpot ecosystems using context engineering and RAG.
- Grounded model responses in internal manufacturing specifications, **reducing hallucinations** and aligning outputs with complex production workflows.

**Full-Stack Shopify Application Suite** | *Preact, GraphQL, SQLite, Shopify API*

- Developed **three custom applications** to automate international tax compliance and optimize checkout logic.
- Achieved a 2x increase in transaction speed and unlocked **\$100K+ in incremental revenue** through reduced cart abandonment

**Enterprise Integration Middleware** | *Node.js, Java, REST APIs, PostgreSQL*

- Engineered a bidirectional sync service between HubSpot CRM and Fishbowl Inventory, automating SKU mapping, polling, and auth.
- Eliminated manual order entry for the sales team, **ensuring 100% data fidelity** and real-time inventory visibility across the organization.

**Order Comparison & Reconciliation Pipeline** | *Python, Streamlit, Pandas*

- Built a reconciliation tool to audit Fishbowl vs. Shopify data, utilizing confidence scoring for SKU matching and automated discrepancy reporting.
- **Reduced fulfillment errors by 75%** by surfacing product configuration, price, and quantity mismatches before orders reached the shipping dock.

**Automated Factory Floor Labeling Ecosystem** | *TypeScript, Electron, ODBC, SQL*

- Engineered a cross-platform desktop application TypeScript/Electron app to automate labeling, **reclaiming \$120K in annual losses** from shipping errors.
- Architected a robust data layer using ODBC to interface directly with MySQL production databases, ensuring sub-second label generation.

**LLM Content Detection Pipeline** | *Python, Transformers, PyTorch*

- Applied prompt engineering and logit bias methods to watermark GPT-3.5 outputs; **fine-tuned a RoBERTa-based classifier** to detect watermarks.
- Achieved an **86% test-set F1 score** in a model-agnostic setting, evaluating the resilience of watermarking against adversarial edits.

## CERTIFICATES

**Deep Learning Specialization** | *DeepLearning.AI*

- Building Convolutional Neural Networks, Recurrent Neural Networks, and Transformers, and applying Transfer & Multi-task learning with TensorFlow

**Google Cybersecurity Professional Certificate** | *Google*

- OWASP Top 10, SIEM with Splunk, intrusion detection, and incident response

**Microsoft Azure Fundamentals** | *Microsoft*

- Provisioning, monitoring, and auto-scaling Azure VMs, Azure Cosmos DB, Azure Containers; securing cloud resources with IAM & RBAC

**DevOps on AWS Specialization** | *Amazon Web Services*

- Infrastructure provisioning with IaC with AWS CloudFormation, AWS ECS & EKS, AWS CloudWatch, and Amazon DynamoDB